

CHAN-HO, IM

Team Leader, DX Innovation Department

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EXPERIENCE

Team Leader, DX Innovation Dept.

Cosmecca Korea Co., Ltd.

 Jun 2016 – Present  Eumseong-gun, Chungcheongbuk-do

- Overseeing company-wide DX & AX initiatives, on-site improvement activities (TPS), and cost reduction efforts.
- Leading 7 Task Force Teams (TFT) across departments to drive manufacturing innovation and digital transformation.
- Achieved a 30% increase in productivity and a 20% reduction in customer complaints.
- Realized 8 billion KRW in cost savings for fiscal year 2025.

CAPSTONE DESIGN PROJECT

YOLO-based Video AI for Automated Line-of-Balance (LOB) Analysis in Manufacturing Packaging Processes

Sungkyunkwan University – Dept. of Smart Factory Convergence

 Mar 2026 – Jun 2026  Team Leader (Team 5)

Summary. YOLOv11-based object detection identifies four classes (Person, Hand, Hat, Product) on a frame-by-frame basis, and SORT/DeepSORT maintains per-worker Track IDs across frames. A GRU-based time-series action recognition model classifies four action types (Pick, Place, Working, Other), and Handling Time & Cycle Time are derived from RoI entry/exit events. Reliability is secured via $\pm\Delta t$ time-window matching and track-level debouncing. These values are automatically substituted into the LOB formula and visualized through a real-time dashboard.

Key Results.

- Detection: mAP@0.5 = 85.0%, Precision = 87.6%, Recall = 79.0%, F1 = 83.0%.
- LOB agreement with manual measurement: **95.0%**.
- Enables automatic conversion of unstructured video into structured productivity metrics (LOB).

GOVERNMENT-SUPPORTED PROJECTS

- 2026 Manufacturing Robot Revitalization Project
- 2025 Manufacturing AI Specialization Project
- 2024 Digital Twin & Safety Enhancement Project
- 2023 Manufacturing Robot Plus Project
- 2022 K-Smart Lighthouse Factory
- 2021 Smart Factory Advancement Project
- 2020 Smart Factory Foundation Project

All projects supported by the Ministry of SMEs and Startups (MSS), Republic of Korea.

REVIEWED LITERATURE

Selected references reviewed by Team 5 during the capstone project on Video AI-based LOB analysis.

MY LIFE PHILOSOPHY

“Continuous improvement is not a destination, but a way of thinking and living.”

RESEARCH OUTCOMES

Patent

Real-time LOB Computation Program based on YOLO and GRU-driven Time-Series Action Recognition.



Software Registration

Real-time LOB Computation Program based on YOLO and GRU-driven Time-Series Action Recognition.



Publication

A Study on Worker Detection and Automated Line-of-Balance (LOB) Analysis in Manufacturing Packaging Lines Using YOLO-Based Video AI.

MOST PROUD OF



Presidential Prize

Awarded the Presidential Prize at the Manufacturing Innovation Awards for improving productivity through Smart Factory establishment.



8B KRW Cost Savings

Realized 8 billion KRW in cost savings through DX and process innovation initiatives in FY2025.



30% Productivity Gain

Achieved a 30% productivity increase and 20% reduction in customer complaints across manufacturing operations.

STRENGTHS

Smart Factory Planning

AI/DX Transformation

Process Optimization

Computer Vision

YOLO / GRU

Team Leadership

Manufacturing Innovation

1. YOLO-DS: Fine-Grained Feature Decoupling via Dual-Statistic Synergy Operator for Object Detection.
2. Detection of Anomalous Behavior of Manufacturing Workers Using Deep Learning-Based Recognition of Human-Object Interaction.
3. SMA-YOLO: A Multi-Scale Object Detection Algorithm for UAV Remote Sensing Images.
4. YOLOv11: An Overview of the Key Architectural Enhancements.
5. A Deep Context Learning-Based PCB Defect Detection Model with Anomalous Trend Alarming System.
6. LiVOS: Light Video Object Segmentation with Gated Linear Matching.
7. Towards Adaptive Human-Centric Video Anomaly Detection.

LANGUAGES

Korean



English



EDUCATION

Graduate Studies

Sungkyunkwan University

Department of Smart Factory Convergence
Capstone Team Leader (Team 5)

CONTACT

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